LUNAR VISIBILITY PROJECT

Mathematical Foundations and Derivations

From lunar coordinates to terrain-aware Sun and Earth visibility

English team reference

Static observer model

This document consolidates the formulas, step-by-step derivations, numerical examples, and mathematical checks for a lunar visibility algorithm. It begins with the local east, north, and up basis, then derives terrain horizons, finite-disk visibility, event times, and operational durations.

The emphasis is on understanding why each expression is valid and where a model assumption or numerical approximation enters. No programming code is included.

Model scope

A fixed site and a fixed radial instrument height; a Moon-fixed reference frame; a radial terrain model; direct geometric visibility of the Sun and Earth. Numerical examples are illustrative, not predictions for a particular landing site or mission.

Version 1.0 | 6 November 2026

Contents

Purpose, assumptions, and notation 1

From latitude and longitude to Cartesian position 1

Deriving the local up direction 2

Deriving the local east direction 3

Deriving the local north direction 4

Checking the basis and its limitations 4

From basis vectors to the transformation matrix 5

Observer position and celestial directions 6

Deriving elevation and azimuth 7

Constructing the terrain skyline 8

Exact terrain geometry and lunar curvature 9

Why the horizon can be below zero elevation 11

Deriving a defensible terrain search radius 12

Deriving the angular size of the Sun and Earth 13

Deriving the visible disk fraction by integration 13

Irregular horizons and overlapping obstructions 15

From visibility to solar-panel power 16

Deriving rise and set times 17

Durations, blackout periods, and simultaneous windows 18

Mathematical checks and implementation boundaries 20

References and attribution 21

Purpose, assumptions, and notation

The east, north, and up expressions are geometric identities under a specified coordinate convention. They follow from spherical coordinates, tangent vectors, and orthogonal projections. The terrain and visibility formulas then follow from the same geometry.

Three kinds of statements are distinguished throughout this document:

- **Exact geometric identities**, valid within the stated model.
- **Physical model assumptions**, such as radial instrument height or a spherical apparent limb.
- **Numerical approximations**, such as finite terrain sampling and linear interpolation in time.

The Moon-fixed Cartesian frame has its origin at the lunar center. Its Z axis points toward the north pole of the selected frame, its X axis passes through the equator and prime meridian, and its Y axis passes through the equator at 90° east longitude.

We use planetocentric latitude φ, east-positive longitude λ, and radial distance r. Latitude is the angle between the radius vector and the equatorial plane. Longitude increases from $+X$ toward $+Y$. These conventions agree with the latitudinal coordinates defined by NAIF's `latrec` routine. [1]

The site coordinates are φ_0, λ_0. The reference radius is R, terrain elevation above that reference sphere is $z(\varphi, \lambda)$, and the fixed instrument height above the local surface is h. Both z and h are radial in this model. The scalar z for terrain elevation should not be confused with the Cartesian component Z.

The local unit vectors are $\mathbf{e}, \mathbf{n}, \mathbf{u}$; local vector components are E, N, U. The celestial azimuth is A, elevation is ε, terrain angular elevation is β, and the skyline is $H(A)$. A central angle along a terrain profile is denoted by ψ. The apparent angular radius of a celestial disk is ρ.

The terrain is assumed to be a continuous, single-valued radial surface bounding an opaque solid body. This is a height-field model. Spherical coordinates can describe a nonspherical surface through the varying radius $R + z$; using them does not discard topography.

All differentiation and trigonometric computation use radians:

$$\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}.$$

Distances must use one consistent unit. Numerical examples often use meters, while SPICE position results normally use kilometers.

From latitude and longitude to Cartesian position

Consider a point at radial distance r and latitude φ. Its projection onto the equatorial plane has length

$$L = r \cos \varphi,$$

and its component perpendicular to that plane is

$$Z = r \sin \varphi.$$

The projected vector makes angle λ with the X axis. Resolving it into its two Cartesian components gives

$$X = L \cos \lambda, \qquad Y = L \sin \lambda.$$

Substituting the expression for L yields

$$\boxed{\mathbf{r}(r,\varphi,\lambda) = \begin{pmatrix} r\cos\varphi\cos\lambda \\ r\cos\varphi\sin\lambda \\ r\sin\varphi \end{pmatrix}.}$$

Each factor has a direct meaning. The radius sets the scale; $\cos\varphi$ selects the equatorial projection; $\cos\lambda$ and $\sin\lambda$ resolve that projection along X and Y; and $\sin\varphi$ gives the vertical component.

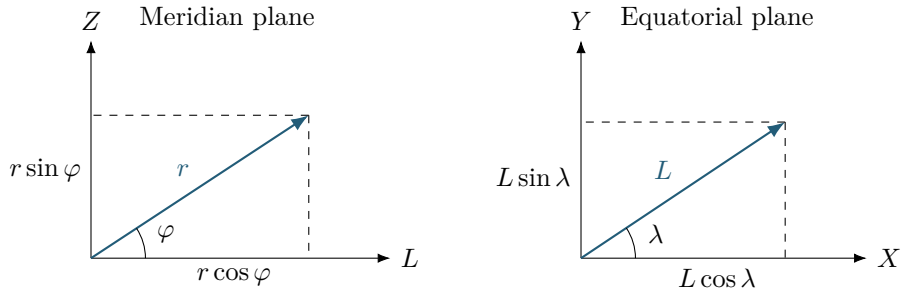


Figure 1: The Cartesian position formula is obtained by two successive orthogonal projections. The sketches are schematic.

Checks at familiar locations

At the equator and prime meridian,

$$\varphi = 0, \quad \lambda = 0 \quad \Longrightarrow \quad \mathbf{r} = \begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix}.$$

At the equator and 90° east longitude,

$$\mathbf{r} = \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix}.$$

At the north pole,

$$\varphi = 90^\circ \quad \Longrightarrow \quad \mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix}.$$

These results agree with the definitions of the Cartesian axes.

Deriving the local up direction

In the chosen model, moving upward means increasing r while keeping latitude and longitude fixed. The radius vector already points in that direction, but its length is r. Dividing by its length produces a unit vector:

$$\mathbf{u} = \frac{\mathbf{r}}{r}.$$

Therefore,

$$\boxed{\mathbf{u} = \begin{pmatrix} \cos\varphi\cos\lambda \\ \cos\varphi\sin\lambda \\ \sin\varphi \end{pmatrix}.}$$

The same expression follows by differentiating with respect to radius:

$$\mathbf{u} = \frac{\partial \mathbf{r}}{\partial r}.$$

This derivative asks: in which direction does the point move when its distance from the center increases slightly?

At $\varphi = 30^\circ$ and $\lambda = 0^\circ$,

$$\mathbf{u} = \begin{pmatrix} \sqrt{3}/2 \\ 0 \\ 1/2 \end{pmatrix}.$$

Local up has both a positive X component and a positive Z component. It is not generally the global Z direction.

Deriving the local east direction

Increase longitude while holding radius and latitude fixed. The resulting motion follows a parallel toward the east. For a small positive longitude increment,

$$\mathbf{r}(\varphi, \lambda + \Delta\lambda) - \mathbf{r}(\varphi, \lambda) \approx \frac{\partial \mathbf{r}}{\partial \lambda}\Delta\lambda.$$

The derivative therefore points along the eastward tangent. More generally, partial derivatives of a parametrized surface are tangent to its coordinate curves. [2]

Differentiate the three components:

$$\frac{\partial \mathbf{r}}{\partial \lambda} = \begin{pmatrix} -r\cos\varphi\sin\lambda \\ r\cos\varphi\cos\lambda \\ 0 \end{pmatrix} = r\cos\varphi \begin{pmatrix} -\sin\lambda \\ \cos\lambda \\ 0 \end{pmatrix}.$$

The negative sign comes from $d(\cos\lambda)/d\lambda = -\sin\lambda$. The third component is zero because $Z = r\sin\varphi$ does not change along a parallel.

For $-\pi/2 < \varphi < \pi/2$, the derivative has length

$$\left\| \frac{\partial \mathbf{r}}{\partial \lambda} \right\| = r\cos\varphi.$$

Normalizing it gives

$$\boxed{\mathbf{e} = \frac{\partial \mathbf{r}/\partial \lambda}{\|\partial \mathbf{r}/\partial \lambda\|} = \begin{pmatrix} -\sin\lambda \\ \cos\lambda \\ 0 \end{pmatrix}.}$$

Latitude controls the radius of the parallel and thus the displacement per radian of longitude. That scale disappears when the derivative is normalized.

For example,

$$\lambda = 0^\circ \Rightarrow \mathbf{e} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \lambda = 90^\circ \Rightarrow \mathbf{e} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}.$$

From a point on the positive Y axis, further eastward motion initially decreases X, as the second result indicates.

Deriving the local north direction

Increase latitude while holding longitude and radius fixed. This follows a meridian toward the north. The relevant tangent is

$$\frac{\partial \mathbf{r}}{\partial \varphi} = \begin{pmatrix} -r\sin\varphi\cos\lambda \\ -r\sin\varphi\sin\lambda \\ r\cos\varphi \end{pmatrix} = r \begin{pmatrix} -\sin\varphi\cos\lambda \\ -\sin\varphi\sin\lambda \\ \cos\varphi \end{pmatrix}.$$

Its length is r. Dividing by r produces

$$\boxed{\mathbf{n} = \begin{pmatrix} -\sin\varphi\cos\lambda \\ -\sin\varphi\sin\lambda \\ \cos\varphi \end{pmatrix}.}$$

At $\varphi = 30^\circ, \lambda = 0^\circ$,

$$\mathbf{n} = \begin{pmatrix} -1/2 \\ 0 \\ \sqrt{3}/2 \end{pmatrix}.$$

Moving north increases Z while reducing the distance to the rotation axis. At this longitude, X therefore decreases.

The spherical-basis derivation by differentiation and normalization is also presented in the University of Illinois dynamics notes. Their inclination angle is measured from $+Z$; it must be replaced by $\pi/2 - \varphi$ to match the latitude convention used here. [3]

Checking the basis and its limitations

Unit lengths and orthogonality

A local orthonormal basis must satisfy

$$\|\mathbf{e}\| = \|\mathbf{n}\| = \|\mathbf{u}\| = 1,$$

and

$$\mathbf{e}\cdot\mathbf{n} = \mathbf{e}\cdot\mathbf{u} = \mathbf{n}\cdot\mathbf{u} = 0.$$

For east,

$$\|\mathbf{e}\|^2 = \sin^2\lambda + \cos^2\lambda = 1.$$

For north,

$$\begin{aligned} \|\mathbf{n}\|^2 &= \sin^2\varphi\cos^2\lambda + \sin^2\varphi\sin^2\lambda + \cos^2\varphi \\ &= \sin^2\varphi(\cos^2\lambda + \sin^2\lambda) + \cos^2\varphi = 1. \end{aligned}$$

Similarly,

$$\|\mathbf{u}\|^2 = \cos^2\varphi(\cos^2\lambda + \sin^2\lambda) + \sin^2\varphi = 1.$$

The east-north dot product is

$$\begin{aligned} \mathbf{e}\cdot\mathbf{n} &= (-\sin\lambda)(-\sin\varphi\cos\lambda) + \cos\lambda(-\sin\varphi\sin\lambda) \\ &= 0. \end{aligned}$$

The north-up dot product is

$$\mathbf{n} \cdot \mathbf{u} = -\sin\varphi\cos\varphi(\cos^2\lambda + \sin^2\lambda) + \sin\varphi\cos\varphi = 0.$$

The same cancellation gives $\mathbf{e} \cdot \mathbf{u} = 0$. Finally,

$$\boxed{\mathbf{e} \times \mathbf{n} = \mathbf{u}.}$$

The basis is therefore orthonormal and right-handed. The three expressions agree with equation (4) in ESA Navipedia's ENU transformation reference. [4]

The exact poles

At $\varphi = \pm\pi/2$, the derivative with respect to longitude vanishes because all meridians meet at the same point. East and north require a chosen reference meridian at the exact pole. After making that choice, the displayed formulas define a consistent local orientation. This is a coordinate singularity, not a physical failure of the visibility geometry.

Radial up is different from a terrain normal

The derivation differentiated a reference sphere at fixed radius. For actual terrain,

$$\mathbf{q} = r(\varphi, \lambda)\mathbf{u}(\varphi, \lambda),$$

the product rule gives

$$\frac{\partial \mathbf{q}}{\partial \varphi} = \frac{\partial r}{\partial \varphi}\mathbf{u} + r\mathbf{n},$$

and

$$\frac{\partial \mathbf{q}}{\partial \lambda} = \frac{\partial r}{\partial \lambda}\mathbf{u} + r\cos\varphi\,\mathbf{e}.$$

The additional terms describe terrain slope. Normalizing these derivatives does not recover the same reference-sphere ENU basis.

Radial up, the terrain-surface normal, and a solar-panel normal are distinct directions. Likewise, planetocentric latitude with radial height must not be mixed with geodetic latitude and height measured along an ellipsoid normal. ESA explicitly distinguishes the spherical and ellipsoidal interpretations. [4]

From basis vectors to the transformation matrix

Each displayed column vector is one local axis expressed in global XYZ components. A matrix is obtained by assembling those axes.

Any displacement vector can be expanded as

$$\mathbf{v} = E\mathbf{e} + N\mathbf{n} + U\mathbf{u}.$$

Let

$$\mathbf{B} = [\,\mathbf{e}\ \mathbf{n}\ \mathbf{u}\,] = \begin{pmatrix} -\sin\lambda & -\sin\varphi\cos\lambda & \cos\varphi\cos\lambda \\ \cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi\sin\lambda \\ 0 & \cos\varphi & \sin\varphi \end{pmatrix}.$$

Then

$$\boxed{\mathbf{v} = \mathbf{B}\begin{pmatrix} E \\ N \\ U \end{pmatrix}.}$$

Thus, a matrix with the local basis vectors as *columns* transforms local ENU components into global XYZ components.

Why the reverse transform uses rows

Taking the dot product with east gives

$$\begin{aligned} \mathbf{v}\cdot\mathbf{e} &= (E\mathbf{e} + N\mathbf{n} + U\mathbf{u})\cdot\mathbf{e} \\ &= E(\mathbf{e}\cdot\mathbf{e}) + N(\mathbf{n}\cdot\mathbf{e}) + U(\mathbf{u}\cdot\mathbf{e}) \\ &= E. \end{aligned}$$

Similarly,

$$N = \mathbf{v}\cdot\mathbf{n}, \qquad U = \mathbf{v}\cdot\mathbf{u}.$$

Therefore,

$$\boxed{\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda & \cos\lambda & 0 \\ -\sin\varphi\cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi \\ \cos\varphi\cos\lambda & \cos\varphi\sin\lambda & \sin\varphi \end{pmatrix}\mathbf{v}.}$$

Now the basis vectors are *rows*: each row computes one dot product. Because the basis is orthonormal,

$$\mathbf{B}^{\mathsf{T}}\mathbf{B} = \mathbf{I}, \qquad \boxed{\mathbf{B}^{-1} = \mathbf{B}^{\mathsf{T}}}.$$

Its determinant is $+1$, consistent with a proper rotation and a right-handed basis.

Observer position and celestial directions

Let $z_0 = z(\varphi_0, \lambda_0)$. If instrument height h is radial, the observer position is

$$r_o = R + z_0 + h, \qquad \boxed{\mathbf{p} = (R + z_0 + h)\mathbf{u}_0.}$$

The site basis is evaluated at φ_0, λ_0.

For example,

$$R = 1\,737\,400 \text{ m}, \qquad z_0 = 1000 \text{ m}, \qquad h = 2 \text{ m}$$

gives

$$r_o = 1\,738\,402 \text{ m}.$$

The reference radius 1 737 400 m matches the sphere used by the cited LOLA polar products. [5]

If height were instead measured along a local terrain normal $\boldsymbol{\nu}$, the position would be $\mathbf{p} = \mathbf{q}_0 + h\boldsymbol{\nu}$. The radial formula would then be inappropriate without further assumptions.

For an uncorrected geometric calculation, let $\mathbf{r}_B(t)$ be the position of a celestial center relative to the lunar center in the same Moon-fixed frame and at the same epoch. The observer-to-body vector is

$$\boxed{\mathbf{v}_B(t) = \mathbf{r}_B(t) - \mathbf{p}.}$$

Subtracting the observer position is essential. A vector from the lunar center is not a vector from the landing site.

For apparent positions, light-time and aberration corrections should be handled consistently by SPICE. NAIF's `spkcpo` supports an observer with constant position in a specified frame and can return a corrected target vector directly relative to that observer. Its returned position vector must not have the site position subtracted a second time. [6]

After obtaining the appropriate observer-relative vector, project it onto the site basis:

$$E = \mathbf{v}_B \cdot \mathbf{e}, \qquad N = \mathbf{v}_B \cdot \mathbf{n}, \qquad U = \mathbf{v}_B \cdot \mathbf{u}.$$

Ephemerides supply the celestial motion and lunar orientation. The local coordinate derivation does not itself predict the orbits.

Deriving elevation and azimuth

Elevation above the mathematical horizon

The horizontal projection of the observer-to-body vector has length

$$L = \sqrt{E^2 + N^2}.$$

The horizontal and vertical components form a right triangle, so

$$\tan \varepsilon = \frac{U}{L}.$$

A quadrant-aware expression is

$$\boxed{\varepsilon = \operatorname{atan2}\left(U, \sqrt{E^2 + N^2}\right).}$$

Positive elevation is above the mathematical horizontal plane; negative elevation is below it.

The same triangle gives

$$\sin \varepsilon = \frac{U}{\sqrt{E^2 + N^2 + U^2}},$$

and hence

$$\boxed{\varepsilon = \arcsin\left(\frac{U}{\|\mathbf{v}_B\|}\right).}$$

These are equivalent for a nonzero vector. The inverse-sine argument may require clipping to $[-1, 1]$ to handle floating-point roundoff. This is the mathematical core of an `elevation()` function.

Azimuth measured from north through east

Adopt 0° for north, 90° for east, 180° for south, and 270° for west. Then

$$N = L \cos A, \qquad E = L \sin A.$$

Therefore,

$$\boxed{A = \operatorname{atan2}(E, N) \pmod{2\pi}.}$$

The argument order is intentional: east first, north second. An ordinary $\arctan(E/N)$ loses quadrant information; $(E, N) = (1, 1)$ and $(-1, -1)$ have the same ratio but point in opposite directions.

When $E = N = 0$, the direction is exactly at the zenith or nadir and azimuth is undefined. If the entire vector is zero, neither angle is defined.

A complete numerical example

At $\varphi_0 = 30^\circ, \lambda_0 = 0^\circ$,

$$\mathbf{e} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{n} = \begin{pmatrix} -1/2 \\ 0 \\ \sqrt{3}/2 \end{pmatrix}, \qquad \mathbf{u} = \begin{pmatrix} \sqrt{3}/2 \\ 0 \\ 1/2 \end{pmatrix}.$$

Take the illustrative displacement vector

$$\mathbf{v} = \begin{pmatrix} \sqrt{3}/2 \\ 1 \\ 5/2 \end{pmatrix}.$$

Its overall scale is irrelevant to the angles. Projection gives

$$E = 1,$$

$$N = -\frac{1}{2}\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\frac{5}{2} = \sqrt{3},$$

$$U = \frac{\sqrt{3}}{2}\frac{\sqrt{3}}{2} + \frac{1}{2}\frac{5}{2} = 2.$$

Thus,

$$L = \sqrt{1 + 3} = 2,$$

$$\boxed{\varepsilon = \operatorname{atan2}(2, 2) = 45^\circ, \qquad A = \operatorname{atan2}(1, \sqrt{3}) = 30^\circ.}$$

The object is 30° east of north and 45° above the mathematical horizon. Reconstructing the vector as $E\mathbf{e} + N\mathbf{n} + U\mathbf{u}$ returns the original XYZ components.

Constructing the terrain skyline

A terrain point at latitude φ, longitude λ, and radial elevation z has position

$$\mathbf{q} = (R + z) \begin{pmatrix} \cos\varphi \cos\lambda \\ \cos\varphi \sin\lambda \\ \sin\varphi \end{pmatrix}.$$

Its observer-relative displacement is

$$\mathbf{v}_T = \mathbf{q} - \mathbf{p}.$$

The same ENU projections and angular formulas give the point's azimuth and angular elevation β.

At a fixed azimuth, the skyline is the largest angular elevation reached by the surrounding surface:

$$\boxed{H(A) = \sup\{\beta \text{ of terrain points in direction } A\}.}$$

The supremum is the least upper bound. When an actual highest angular point exists, it is the ordinary maximum. This distinction matters for an observer exactly on a smooth surface.

For example, if successive ridges in one direction have angular elevations 1°, 3°, and 2°, then $H(A) = 3°$. A point-like Sun at 2.5° is blocked; one at 3.5° is clear.

For distant targets such as the Sun and Earth, which lie beyond all lunar terrain, define the signed clearance

$$\boxed{g(t) = \varepsilon(t) - H(A(t)).}$$

A positive clearance means the direction is clear; a negative clearance means it is blocked. Exact equality is a grazing boundary case and needs an explicit convention. For a nearby target in front of an obstacle, one must test the finite observer-to-target segment instead of using the distant-target skyline alone.

Because the site, instrument height, and terrain are static in the Moon-fixed frame, $H(A)$ can be precomputed and reused for different dates. A maximum over sampled pixels or azimuth bins is an approximation to the continuous skyline; narrow blocking peaks can be missed.

Exact terrain geometry and lunar curvature

A terrain profile at a fixed azimuth

The unit horizontal direction for azimuth A is

$$\boxed{\mathbf{w}(A) = \cos A\,\mathbf{n} + \sin A\,\mathbf{e}.}$$

It equals north at $A = 0°$ and east at $A = 90°$.

Let ψ be the central angle between the site radius and the radius of a terrain point along this direction. The corresponding unit radius vector is

$$\boxed{\mathbf{u}_{\psi,A} = \cos\psi\,\mathbf{u} + \sin\psi\,\mathbf{w}(A).}$$

This traces a unit circle in the plane spanned by two perpendicular unit vectors. Its norm is one, and $\mathbf{u}_{\psi,A} \cdot \mathbf{u} = \cos\psi$, confirming the central angle.

If the surface radius at that location is $r(\psi, A)$,

$$\mathbf{q} = r(\psi, A)\left[\cos\psi\,\mathbf{u} + \sin\psi\,\mathbf{w}(A)\right].$$

Subtract $\mathbf{p} = r_o\mathbf{u}$:

$$\mathbf{q} - \mathbf{p} = (r\cos\psi - r_o)\mathbf{u} + r\sin\psi\,\mathbf{w}(A).$$

The first coefficient is the vertical component. For $0 < \psi < \pi$, the second is the nonnegative horizontal length. Therefore,

$$\boxed{\beta(\psi, A) = \operatorname{atan2}\left(r(\psi, A)\cos\psi - r_o,\ r(\psi, A)\sin\psi\right).}$$

This is exact within the radial surface model. Lunar curvature is already included in the sine and cosine terms.

The profile follows a great circle on the reference sphere. On a polar stereographic DEM, its projected path is generally curved. Also, $R\psi$ is reference-sphere arc length, not distance along the uneven terrain or Euclidean distance on the map.

Deriving the small-distance curvature correction

For small ψ,

$$\cos\psi \approx 1 - \frac{\psi^2}{2}, \qquad \sin\psi \approx \psi.$$

With $r = R + z$ and $r_o = R + z_0 + h$,

$$\begin{aligned} U &= (R+z)\cos\psi - (R + z_0 + h) \\ &\approx z - z_0 - h - \frac{(R+z)\psi^2}{2}. \end{aligned}$$

If heights are small compared with R, and $d = R\psi$,

$$U \approx z - z_0 - h - \frac{d^2}{2R}, \qquad L \approx d.$$

Hence,

$$\boxed{\beta \approx \operatorname{atan2}\left(z - z_0 - h - \frac{d^2}{2R},\ d\right).}$$

The term $d^2/(2R)$ is the approximate drop of the spherical surface below the site's tangent plane.

Numerical example: a distant ridge

Use

$$R = 1\,737\,400 \text{ m}, \quad z_0 = 0, \quad h = 2 \text{ m},$$

and place an illustrative ridge with $z = 1000$ m at reference-sphere arc distance $d = 20\,000$ m. Then

$$\psi = \frac{20\,000}{1\,737\,400}.$$

Ignoring curvature gives

$$\arctan\frac{1000 - 2}{20\,000} \approx 2.85669^\circ.$$

The exact spherical expression gives

$$\boxed{\beta \approx 2.52606^\circ.}$$

The approximate surface drop is

$$\frac{20\,000^2}{2 \cdot 1\,737\,400} \approx 115.1 \text{ m}.$$

Do not subtract this drop again when using exact XYZ positions or the exact sine-cosine formula. That would count curvature twice.

For close terrain, the exact numerator can also be written as

$$r\cos\psi - r_o = (r - r_o) - 2r\sin^2(\psi/2),$$

which reduces cancellation between two large, nearly equal radii.

Why the horizon can be below zero elevation

Consider a smooth sphere of radius R and an observer at height $h > 0$. The limiting line of sight is tangent to the sphere. The radius to the tangent point is perpendicular to that line, creating a right triangle with hypotenuse $R + h$ and one leg R.

The central angle to the tangent point therefore satisfies

$$\cos \psi_h = \frac{R}{R+h}, \qquad \psi_h = \arccos \frac{R}{R+h}.$$

The tangent sightline is depressed below the observer's local horizontal by the same angle:

$$\boxed{H_{\text{sphere}} = -\arccos \frac{R}{R+h}.}$$

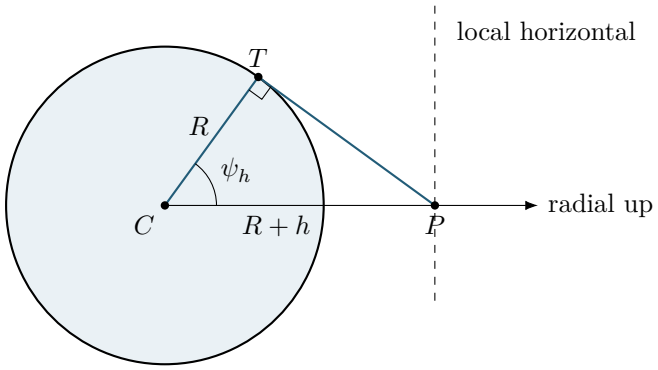


Figure 2: A tangent sightline determines the smooth-sphere horizon. C is the lunar center, P the observer, and T the tangent point. Observer height is exaggerated.

For $R = 1\,737\,400$ m and $h = 2$ m,

$$\boxed{H_{\text{sphere}} \approx -0.0869366^\circ,} \qquad R\psi_h \approx 2636.21 \text{ m}.$$

A point-like source at elevation -0.05° remains visible in this smooth-sphere example. Clamping all negative skyline elevations to zero would give an incorrect result.

The limiting case of zero instrument height

For a smooth sphere and $h = 0$,

$$\beta(\psi) = \operatorname{atan2}(\cos \psi - 1, \sin \psi) = -\frac{\psi}{2}, \quad 0 < \psi < \pi.$$

Every distinct surface point has negative elevation, but

$$\boxed{\sup_{\psi > 0} \beta(\psi) = 0.}$$

The supremum is approached as $\psi \to 0$ and is not attained by a distinct point. This explains why the continuous skyline should not always be defined as an attained maximum.

Exclude the observer point itself, where the displacement vector vanishes. For differentiable radial terrain at $h = 0$, the one-sided near-site limit is

$$\lim_{\psi \to 0+} \beta(\psi, A) = \arctan\left(\frac{1}{r_0} \left. \frac{\partial r(\psi, A)}{\partial \psi} \right|_{\psi = 0^+} \right),$$

where r_0 is the local surface radius. It follows by expanding both numerator and denominator to first order. Removing nearby terrain samples without accounting for this limit can erase the local slope from the horizon model.

At exact equality $\varepsilon = H(A)$, grazing contact must be treated explicitly. A strict positive-clearance rule is a useful operational convention, but the equality case is not resolved by the supremum alone in every geometry.

Deriving a defensible terrain search radius

Assume the opaque body contains a sphere of radius $r_{\min} > 0$, while its entire surface lies inside a sphere of radius $r_{\max}$:

$$r_{\min} \le r(\varphi, \lambda) \le r_{\max}, \qquad r_o \ge r_{\min}.$$

These must be guaranteed bounds for the model, not just extrema inside an arbitrarily cropped DEM.

Draw a tangent from the observer to the inner sphere. The corresponding central angle is

$$\alpha = \arccos \frac{r_{\min}}{r_o}.$$

Draw another tangent from a terrain point of radius r. Its central angle is

$$\eta(r) = \arccos \frac{r_{\min}}{r} \le \arccos \frac{r_{\min}}{r_{\max}}.$$

The limiting observer-to-terrain segment just touches the inner sphere when the separation angle is $\alpha + \eta(r)$. At a greater separation, that segment intersects the opaque inner sphere.

Consequently, no terrain beyond

$$\boxed{\psi_{\text{cut}} = \arccos \frac{r_{\min}}{r_o} + \arccos \frac{r_{\min}}{r_{\max}}}$$

can raise the visible skyline. Its reference-sphere arc distance is $d_{\text{cut}} = R\psi_{\text{cut}}$.

The inner sphere also supplies the guaranteed skyline floor

$$H_{\text{floor}} = -\alpha.$$

One consistent bounded expression is therefore

$$H(A) = \max \left[-\alpha, \sup_{0 < \psi \le \psi_{\text{cut}}} \beta(\psi, A) \right].$$

If both tangent angles are zero, the profile interval is empty and the limiting result is $H = 0$; do not evaluate a zero-displacement angle.

Illustrative bound

For the hypothetical global bounds

$$r_{\min} = R, \quad r_{\max} = R + 1000 \text{ m}, \quad r_o = R + 2 \text{ m},$$

with $R = 1\,737\,400$ m,

$$\boxed{d_{\text{cut}} \approx 61.57 \text{ km}.}$$

These are invented bounds for a geometric example, not the actual global elevation range of the Moon.

If any terrain lies below the reference sphere, R cannot automatically serve as the opaque inner radius. Use a genuine lower bound. For a triangulated surface, the interior of a flat triangle can also have a smaller radius than its vertices, so a vertex-only minimum is not automatically sufficient.

This search bound is derived here from two tangents. It is not presented as a separately cited NASA formula or as a guarantee about unresolved real terrain.

Deriving the angular size of the Sun and Earth

Let a spherical celestial body have physical radius R_B, and let the distance from the observer to its center be $D_B > R_B$. A sightline toward the apparent limb is tangent to the body. The radius to the tangent point is perpendicular to that sightline, so

$$\sin \rho = \frac{R_B}{D_B}.$$

Therefore,

$$\boxed{\rho = \arcsin \frac{R_B}{D_B}.}$$

This is the apparent angular *radius*; the diameter is 2ρ. The distance is measured from the observer, not from the lunar center.

For rounded illustrative values

$$R_B = 700\,000 \text{ km}, \qquad D_B = 150\,000\,000 \text{ km},$$

the angular radius is approximately

$$\rho \approx 0.2674^\circ.$$

If the disk center is 0.20° below a locally flat horizon, its upper limb still extends above that horizon. Atmospheric refraction is not needed to explain this purely geometric effect.

Let

$$\Delta = \varepsilon - H.$$

For a horizon that is approximately flat across the disk,

$$\Delta \leq -\rho \implies \text{the disk has zero exposed area,}$$

$$\Delta \geq \rho \implies \text{the whole disk is exposed.}$$

Intermediate values correspond to partial exposure. These tests are not sufficient for an arbitrary mountainous skyline across the disk.

Deriving the visible disk fraction by integration

The local planar approximation

Represent the small apparent disk in a local sky plane by a circle of radius ρ. Let x and y be horizontal and vertical offsets from its center:

$$x^2 + y^2 \leq \rho^2.$$

The center is above the horizon by Δ. A disk point is visible when

$$y > -\Delta.$$

At a given height y, the circle extends from $-\sqrt{\rho^2 - y^2}$ to $+\sqrt{\rho^2 - y^2}$. Its horizontal strip width is

$$w(y) = 2\sqrt{\rho^2 - y^2}.$$

Thus a thin strip has area

$$dS = 2\sqrt{\rho^2 - y^2}\, dy.$$

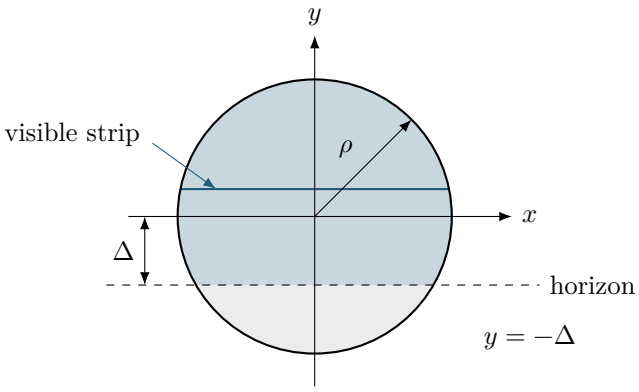


Figure 3: Summing exposed horizontal strips gives the visible circular segment. The example places the center half a radius above the horizon.

For partial exposure, the visible area is

$$S_{\text{vis}} = \int_{-\Delta}^{\rho} 2\sqrt{\rho^2 - y^2}\, dy.$$

The integral literally adds the areas of all visible thin strips.

Evaluating the integral

Use the substitution

$$y = \rho \sin\theta, \qquad dy = \rho \cos\theta\, d\theta.$$

For $-\pi/2 \le \theta \le \pi/2$,

$$2\sqrt{\rho^2 - y^2}\, dy = 2\rho^2 \cos^2\theta\, d\theta.$$

The identity $2\cos^2\theta = 1 + \cos 2\theta$ gives

$$\int 2\rho^2 \cos^2\theta\, d\theta = \rho^2 \left(\theta + \tfrac{1}{2}\sin 2\theta\right).$$

Returning to y,

$$\int 2\sqrt{\rho^2 - y^2}\, dy = y\sqrt{\rho^2 - y^2} + \rho^2 \arcsin(y/\rho).$$

Substitute the limits and divide by the full circle area $\pi\rho^2$. With

$$k = \frac{\Delta}{\rho},$$

the visible area fraction is

$$\boxed{f = \begin{cases} 0, & k \le -1, \\ \dfrac{1}{2} + \dfrac{\arcsin k + k\sqrt{1-k^2}}{\pi}, & -1 < k < 1, \\ 1, & k \ge 1. \end{cases}}$$

Checks and a numerical example

At the midpoint, $k = 0$ gives $f = 1/2$. At half a radius above the horizon,

$$k = \tfrac{1}{2} \quad \Longrightarrow \quad f \approx 0.8045.$$

At half a radius below,

$$k = -\tfrac{1}{2} \quad \Longrightarrow \quad f \approx 0.1955.$$

For

$$\rho = 0.30^\circ, \qquad H = 2^\circ, \qquad \varepsilon = 1.85^\circ,$$

we obtain

$$k = \frac{1.85 - 2}{0.30} = -0.5,$$

so approximately 19.55% of the disk area is exposed.

The area fraction itself is geometric. Interpreting it as a fraction of incident light also requires a brightness model and an appropriate treatment of the receiving surface. The circular-segment expression assumes a small disk in a local plane and a straight horizontal skyline across that disk.

Irregular horizons and overlapping obstructions

Integrating a visibility mask

For a mountainous horizon, each disk point has its own azimuth, elevation, and corresponding terrain obstruction. Define

$$M(x, y) = \begin{cases} 1, & \text{the direction is clear,} \\ 0, & \text{the direction is blocked.} \end{cases}$$

In the same small-disk planar representation,

$$\boxed{f = \frac{1}{\pi\rho^2} \iint_{\text{disk}} M(x, y)\, dx\, dy.}$$

A numerical approximation is

$$\boxed{f \approx \frac{\sum_i w_i M_i}{\sum_i w_i},}$$

where w_i is the area represented by sample i. Equal-area samples allow the simpler fraction of visible samples.

Equal spacing in radius is not equal spacing in area. A disk of half the radius contains only one quarter of the total area:

$$\frac{\pi(\rho/2)^2}{\pi\rho^2} = \frac{1}{4}.$$

For uniform area sampling, a uniform variable $q \in [0, 1]$ therefore maps to sample radius

$$\boxed{r_{\text{sample}} = \rho\sqrt{q},}$$

with an independently uniform polar angle. Here r_{sample} is an offset in the disk plane, not a lunar radial coordinate.

Earth blocking the Sun

Let $\hat{\mathbf{s}}_\odot$ and $\hat{\mathbf{s}}_\oplus$ be unit directions toward the Sun and Earth. Their center separation is

$$\gamma = \arccos\left(\hat{\mathbf{s}}_\odot \cdot \hat{\mathbf{s}}_\oplus\right).$$

For a spherical Earth in front of the Sun, a solar-disk ray in unit direction $\hat{\mathbf{q}}$ lies inside the apparent Earth disk when

$$\hat{\mathbf{q}} \cdot \hat{\mathbf{s}}_\oplus \geq \cos\rho_\oplus.$$

This follows directly from the dot-product definition of the angle between unit vectors. An opaque spherical-Earth model ignores transmission through Earth's atmosphere.

For each individual ray, combine the two clear-direction masks:

$$\boxed{M_{\text{total}} = M_{\text{terrain}} M_{\text{Earth}}.}$$

Integrate the combined mask afterward. In general,

$$f_{\text{total}} \neq f_{\text{terrain}} f_{\text{Earth}}.$$

If both obstructions leave the same half of the Sun open, the result is 50%. If they leave opposite halves open, the result is zero. Multiplying two average fractions of 0.5 would incorrectly give 25% in both cases.

For Earth visibility, define whether the reported criterion concerns its center, any exposed limb, the full disk, or a particular ground station. NASA notes that counting any visible part of Earth's disk can be optimistic for links to specific DSN stations. [5]

From visibility to solar-panel power

The exposed fraction of the solar disk does not by itself specify electrical power. Let S_p be panel area, η conversion efficiency, and I_0 the unobstructed direct-normal irradiance at the observer's distance. Let $\mathbf{m}$ be the panel's unit normal and $\hat{\mathbf{s}}$ the unit direction from the observer toward the solar center.

If θ is the incidence angle,

$$\mathbf{m} \cdot \hat{\mathbf{s}} = \cos\theta.$$

For front-side incidence, $0 \leq \theta \leq \pi/2$, the panel's projected area is

$$S_{\text{effective}} = S_p \cos\theta.$$

For a uniformly bright small disk, when variation of incidence angle across the disk is negligible,

$$\boxed{P \approx \eta S_p I_0 f \max(0, \mathbf{m} \cdot \hat{\mathbf{s}}).}$$

At $\theta = 60^\circ$, the effective area is halved. The maximum with zero prevents the front surface from receiving a negative contribution from rays behind it.

The more general radiometric expression integrates each open ray with its own incidence factor:

$$P = \eta S_p \int_{\text{solar disk}} L(\hat{\mathbf{q}}) M(\hat{\mathbf{q}}) \max(0, \mathbf{m} \cdot \hat{\mathbf{q}})\, d\Omega.$$

Here L is radiance and $d\Omega$ is an element of solid angle. In angular coordinates about an axis, $d\Omega = \sin\theta\, d\theta\, d\chi$. The radiance-to-irradiance relation and cosine factor are described in the cited Cambridge radiometry notes. [7]

The planar disk-area fraction, visible solid-angle fraction, and received-power fraction are not exactly identical quantities. The small-disk approximation is useful, but it becomes less reliable for a panel illuminated at a grazing angle.

Deriving rise and set times

For a point target, define

$$g(t) = \varepsilon(t) - H(A(t)).$$

A transition occurs at a root

$$g(t_*) = 0.$$

Suppose two neighboring times have values $g_0 = g(t_0)$ and $g_1 = g(t_1)$. Over a sufficiently short interval, linear interpolation assumes

$$g(t) \approx g_0 + (g_1 - g_0)\frac{t - t_0}{t_1 - t_0}.$$

Set this to zero and solve for time:

$$\boxed{t_* \approx t_0 - \frac{g_0}{g_1 - g_0}(t_1 - t_0).}$$

This is the basis of a linearly interpolated `next_event()` estimate when an appropriate transition bracket has been found.

For example,

$$g(12{:}00) = -0.2^\circ, \qquad g(13{:}00) = +0.4^\circ.$$

The increase is 0.6° per hour; reaching zero requires one third of that increase:

$$\frac{0.2}{0.6} \cdot 60 \text{ min} = 20 \text{ min}.$$

The interpolated transition is therefore approximately 12:20.

Event detection and boundary refinement

Linear interpolation is an approximation, not an exact law of celestial motion. With complex terrain, the skyline can change slope as different ridges define the boundary. Recompute the geometry inside the bracket to refine the crossing.

No sign change between hourly endpoints does not prove that no event occurred. A short light window may open and close between those endpoints. NAIF distinguishes the search step, which

affects event completeness, from the convergence tolerance, which controls refinement of detected boundaries. Actual accuracy also depends on the input data. [8]

For a required exposed fraction $f_{\min} > 0$, use

$$g(t) = f(t) - f_{\min}.$$

For example, $f_{\min} = 0.8$ means at least 80% of the disk must be exposed.

For continuous f, signed-root searches apply to thresholds $0 < f_{\min} < 1$. At full exposure, $f - 1$ reaches a zero plateau rather than changing sign. A fraction estimated by counting finitely many rays is also a staircase function. For those cases, detect and refine changes of the Boolean condition $f \geq f_{\min}$ instead of requiring a continuous sign-changing root.

The first appearance of any light needs care: $f(t)$ never becomes negative, so searching for a sign change in f alone is unsuitable. For a flat skyline, a signed first-limb clearance is

$$g_{\text{first}}(t) = \varepsilon(t) + \rho(t) - H(A(t)).$$

For an irregular skyline, use the maximum clearance over the disk directions or a clearly specified positive exposed-fraction threshold. These clearance expressions describe emergence above terrain in the absence of other obstructions. When Earth occultation is included, determine the onset of direct sunlight from the combined visibility mask.

How angular uncertainty affects timing

Near a simple crossing with $g'(t_*) \neq 0$, a first-order expansion gives

$$\delta g \approx g'(t_*)\,\delta t.$$

Thus the magnitude of the timing uncertainty is approximately

$$\boxed{|\delta t| \approx \frac{|\delta g|}{|g'(t_*)|}.}$$

An angular uncertainty of 0.03° at a clearance-change rate of 0.01° per minute corresponds to about three minutes. Refining the numerical root to one second does not remove that input uncertainty. Near a tangential crossing, this first-order estimate becomes ill-conditioned.

Durations, blackout periods, and simultaneous windows

Define a binary state using an explicit illumination criterion:

$$I_S(t) = \begin{cases} 1, & \text{the selected illumination criterion is satisfied,} \\ 0, & \text{otherwise.} \end{cases}$$

Total usable illumination time is

$$\boxed{T_S = \int_{t_0}^{t_1} I_S(t)\,dt.}$$

If the separate light windows have refined boundaries $[a_k, b_k]$, the same quantity is

$$\boxed{T_S = \sum_k (b_k - a_k).}$$

The longest uninterrupted window is

$$\boxed{T_{\max} = \max_k (b_k - a_k).}$$

If there are no qualifying windows within the analysis interval, define $T_{\max} = 0$. For insufficient illumination or Earth blackout, apply the same interval calculation to the corresponding complementary state. Complete darkness is specifically $f = 0$; it is not the same as falling below a positive operating threshold. If the criterion is $f \geq f_{\min}$, require $0 < f_{\min} \leq 1$. For any exposed sunlight, use $f > 0$.

A `longest_run()` calculation over equally spaced Boolean samples is an estimate of these intervals. Three positive observations at 12:00, 13:00, and 14:00 span two hours from first to last sample. The actual window may begin before 12:00 and end after 14:00. A count multiplied by the sampling interval needs an explicit convention about the intervals represented by samples.

Equivalent full-disk exposure and energy

The equivalent duration of complete disk exposure is

$$\boxed{T_{\text{eq}} = \int_{t_0}^{t_1} f(t)\, dt.}$$

If the disk is completely exposed for two hours and half exposed for another hour,

$$T_{\text{eq}} = 2 + 0.5 = 2.5 \text{ h}.$$

Some sunlight is present for three hours, but the equivalent full-disk exposure is only two and a half hours. This quantity has units of time. Electrical energy instead requires

$$\boxed{W = \int P(t)\, dt.}$$

With power in watts, integrating over seconds gives joules; integrating over hours gives watt-hours. For nonuniform time samples, a trapezoidal approximation to equivalent exposure is

$$T_{\text{eq}} \approx \sum_i \frac{f_i + f_{i+1}}{2}(t_{i+1} - t_i).$$

Binary windows are preferably summed using refined transition boundaries.

Simultaneous sunlight and Earth visibility

Let $I_E(t)$ indicate whether the selected Earth-visibility criterion is satisfied. The simultaneous-window duration is

$$\boxed{T_{\text{joint}} = \int I_S(t) I_E(t)\, dt.}$$

Multiply the states at the same time, not the separate monthly visibility percentages. If both are individually available for half of a period, their overlap can range from zero to half of that period.

If the calculation ends while a state is still active, report that the corresponding transition was not found within the searched interval. Do not infer that it will never occur.

Mathematical checks and implementation boundaries

The coordinate and ENU expressions have been checked algebraically for unit length, mutual orthogonality, and right-handedness. The worked XYZ-to-ENU example reconstructs its original vector and gives $A = 30^\circ$, $\varepsilon = 45^\circ$. The ridge and smooth-sphere examples agree with direct three-dimensional calculations.

The following conditions determine whether a correct formula is being applied to the intended problem:

- Keep global and local frames consistent. Basis columns map ENU to XYZ; basis rows map XYZ to ENU.
- Subtract the observer position only when the supplied target vector is center-relative.
- Keep latitude convention, reference frame, height definition, units, and epoch consistent.
- Separate radial up, terrain normal, and panel normal.
- Count curvature once, and preserve physically valid negative skyline elevations.
- Distinguish the continuous skyline from a discrete estimate. Pixel centers, coarse azimuth bins, and averaged overview layers can miss blocking terrain.
- Use a genuine opaque inner-sphere bound when limiting terrain range, especially where DEM elevations are negative.
- Distinguish center visibility, finite-disk exposure, received irradiance, and electrical energy.
- Separate event discovery from boundary refinement, and report the finite interval over which conclusions were calculated.

These checks establish consistency of the stated geometry. They do not validate the accuracy of a particular DEM, kernel set, coordinate conversion, sampling strategy, or mission prediction. The reference notebook's illustrative terrain must be replaced by real terrain before its skyline can represent a real site.

References and attribution

The mathematical derivations in this document are written out directly from definitions. References below provide coordinate conventions, independent formula checks, the relevant SPICE interfaces, lunar data context, and the radiometric relation. The terrain search-radius argument is an independent geometric derivation and is not attributed to a NASA publication.

[1] NAIF: latitudinal to rectangular coordinates

Jet Propulsion Laboratory, Navigation and Ancillary Information Facility. *latrec_c*. Defines radius, latitude, east-positive longitude, and the conversion to Cartesian coordinates.

https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/cspice/latrec_c.html

[2] University of Alberta: parametric surfaces

Vincent Bouchard, MATH 215, Section 5.4. *Parametric surfaces in* $\mathbb{R}^n$. Explains coordinate partial derivatives as tangent vectors to a parametrized surface.

https://sites.ualberta.ca/~vbouchar/MATH215/section_parametric_surfaces.html

[3] University of Illinois: spherical coordinates

Dynamics: Spherical coordinates. Gives coordinate conversion and spherical basis vectors obtained by differentiation and normalization. Its inclination convention differs from latitude by $\pi/2$.

https://dynamicref.web.engr.illinois.edu/rvs.html

[4] ESA Navipedia: ECEF and ENU transformations

J. Sanz Subirana, J. M. Juan Zornoza, and M. Hernández-Pajares (2011). *Transformations between ECEF and ENU coordinates*. Equation (4) provides the local basis; equation (6) gives the inverse coordinate transformation. The same geometry applies to a lunar body-fixed frame with corresponding conventions.

https://gssc.esa.int/navipedia/index.php/Transformations_between_ECEF_and_ENU_coordinates

[5] NASA GSFC: lunar polar illumination

Erwan Mazarico. *Lunar Polar Illumination*, Planetary Geology, Geophysics and Geochemistry Laboratory. Describes LOLA-based polar products, the 1737.4 km reference sphere, finite-disk illumination and Earth visibility, and relevant interpretation limits.

https://pgda.gsfc.nasa.gov/products/69

[6] NAIF: constant-position observer

Jet Propulsion Laboratory, Navigation and Ancillary Information Facility. *spkcpo_c*. Computes a target state relative to an observer whose position is fixed in a specified reference frame, with supported aberration corrections.

https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/cspice/spkcpo_c.html

[7] University of Cambridge: radiometry

Rafał Mantiuk. *Reflection models and radiometry*, Advanced Graphics, Computer Laboratory. The radiance-to-irradiance relation is presented on slide 10, including the incidence cosine and solid-angle element.

https://www.cl.cam.ac.uk/teaching/1718/AdvGraph/01_Reflection_models.pdf

[8] NAIF: geometric event searches

Jet Propulsion Laboratory, Navigation and Ancillary Information Facility. *gfoclt_c*. Its search-process discussion distinguishes sampling step, boundary-refinement tolerance, and actual accuracy limited by input data.

https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/cspice/gfoclt_c.html

References checked on 6 November 2026.